NPTEL ONLINE CERTIFICATION COURSES

MARKETING ANALYTICS

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Module 02:

WHAT CONSUMERS WANT

Choice-Based Conjoint (CBC)

- Gained popularity in early 90s
- We ask the respondents a set of options and ask them which one they want to choose
- None of these is also an option
- Multiple “held-constant alternatives”

If you were considering buying golf balls for your next outing and these were the only alternatives, which would you choose?

(1 of 14)

Brand:	High-Flyer Pro, by Smith and Forester	High-Flyer Pro, by Smith and Forester	Long Shot, by Performance Plus	
Performance:	Drives 10 yards farther than the average ball	Drives 15 yards farther than the average ball	Drives 15 yards farther than the average ball	None: I Wouldn't Purchase Any of These
Price:	$6.99 for package of 3 balls	$6.99 for package of 3 balls	$10.99 for package of 3 balls	
	Select	Select	Select	Select

https://www.sawtoothsoftware.com/help/lighthouse-studio/manual/hid_web_whatcbc.html

Strengths and Weakness

Strengths	Weaknesses
• Mimic real world • Can investigate interactions, alternative-specific effects • Paper/Computer • None of these • Multiple held constant alternatives	• Larger sample size required • Less attributes is ideal • May lead to heuristic based responses • Complex analysis

Summary Profile of the Respondents

Total number of respondents	303
Number of female respondents	127
Average Age	36.15 years
Average Number of Members in Household	3.64
Average Family Income (Monthly)	USD 167
Average per capita income per day	USD 1.44
Maximum per capita income per day in the sample	USD 1.96
Percentage having health insurance	4%
Profession	33% Low paid service, 23% Business in unorganized sector, 13% Farmer, 12% Daily wage worker/labor/factor worker, 19% Others

Conjoint Attributes

	Level 1	Level 0
Distance	Up to 5km	More than 5 km
Reputation	High	Low
Delivery Method	Telemedicine	Face to face
Payment	Upfront	Not Upfront
Price of the Hospital	Low (Upto Rs. 100 for consultation)	High (Above Rs. 100 for consultation)

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Call:
clogit(choice ~ (Distance + Reputation + Delivery + Payment.Method +
    Price) + strata(V1), data = data)

                 coef exp(coef) se(coef)     z      p
Distance       1.0712    2.9190   0.0596 17.96 <2e-16
Reputation     1.7363    5.6762   0.0771 22.53 <2e-16
Delivery       0.1511    1.1631   0.0772  1.96   0.05
Payment.Method 0.7319    2.0789   0.0690 10.61 <2e-16
Price          1.6307    5.1075   0.0618 26.40 <2e-16

Likelihood ratio test=2451  on 5 df, p=0
n= 9696, number of events= 2424
> AIC(clogout1)
[1] 4279.903
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Direct Effects

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> clogout2
Call:
clogit(choice ~ (Distance + Reputation + Delivery + Payment.Method +
    Price) * (Gender + HealthInsurance) + strata(V1), data = data)

                                  coef exp(coef) se(coef)     z       p
Distance                        1.2752    3.5795   0.0861 14.80 < 2e-16
Reputation                      1.8161    6.1481   0.1086 16.73 < 2e-16
Delivery                        0.0387    1.0394   0.1148  0.34 0.73608
Payment.Method                  0.8221    2.2754   0.1012  8.13 4.4e-16
Price                           1.9967    7.3643   0.0920 21.69 < 2e-16
Gender                              NA        NA   0.0000    NA      NA
HealthInsurance                     NA        NA   0.0000    NA      NA
Distance:Gender                -0.3873    0.6789   0.1209 -3.20 0.00136
Distance:HealthInsurance       -0.0259    0.9744   0.5581 -0.05 0.96293
Reputation:Gender              -0.1453    0.8648   0.1560 -0.93 0.35166
Reputation:HealthInsurance      1.1817    3.2599   1.0439  1.13 0.25765
Delivery:Gender                 0.2178    1.2433   0.1565  1.39 0.16409
Delivery:HealthInsurance        0.4461    1.5622   0.5918  0.75 0.45101
Payment.Method:Gender          -0.1441    0.8658   0.1397 -1.03 0.30220
Payment.Method:HealthInsurance -0.3914    0.6761   0.5418 -0.72 0.47007
Price:Gender                   -0.6778    0.5077   0.1253 -5.41 6.3e-08
Price:HealthInsurance          -1.7111    0.1807   0.4623 -3.70 0.00021

Likelihood ratio test=2503  on 15 df, p=0
n= 9696, number of events= 2424
> AIC(clogout2)
[1] 4247.617
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Additional Results

- Men are less price and distance sensitive
- If health insurance is available, consumers are less price sensitive
- However, they have affinity towards tele-medicine**

- ***Consumers with bank accounts → Less price and distance sensitive, prefers reputed doctors.***
- ***Consumers with local ambulance facility → Less price and distance sensitive, prefer F2F interactions, prefer EMI over upfront payments***

THANK YOU